



What Is a Life Cycle?

Life Science — Life Cycles — Unit 1, Lesson 1 of 6

 **Grade:** 3rd Grade

 **Duration:** 45–55 minutes

 **Subject:** Life Science — Life Cycles

ONE BIG PATTERN

Life cycles can look very different, but they all show the same science idea. A living thing begins life, grows and changes, becomes an adult that can reproduce, and eventually dies.

Birth



Growth



Reproduction



Death

Butterflies, frogs, bean plants, and humans all fit this pattern, even though their stage names and changes are different.

NGSS Standard:

3-LS1-1 — Develop models to describe that organisms have unique and diverse life cycles but all have in common birth, growth, reproduction, and death.

Learning Objective

Students will *compare* several organisms, identify the big pattern all life cycles share, and *build a model* that uses one organism's stages as evidence for birth, growth, reproduction, and death.



The Big Idea



Every living thing has a **life cycle**, but not every life cycle looks the same. A frog, a butterfly, a bean plant, and a human all move through different stages. Even so, they all share the same big idea: **life cycles have a pattern of birth, growth, reproduction, and death.**

This is a preview. Unlock the full lesson.

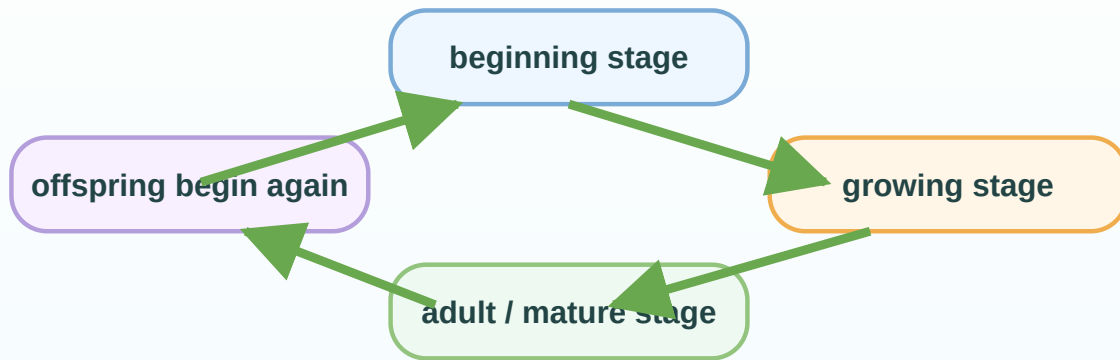
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SCIENCE SNAPSHOT



Life cycles vary in detail, but they all track how living things grow and make more of their kind.

What is happening?

A life cycle is the sequence of stages an organism goes through as it grows, survives, and reproduces. Different organisms look different at each stage, but the big pattern is that living things begin, grow, and eventually make new offspring.

Why it works that way

Life cycles matter because they help scientists compare organisms and understand what each stage is for. Some stages are specialized for feeding, moving, dispersing, or reproducing, so the shape and behavior of the organism may change over time.

Look for this in the lesson

Students should focus on the repeating pattern, not just memorizing stage names. The science becomes deeper when they explain what changes and what stays part of the same cycle.

Ask while you read: How does this organism change across stages, and what big pattern stays the same?



Compare These Life Cycles

Read each row from left to right. Notice what changes, and notice what stays the same about the big pattern.

Butterfly

 Butterfly egg on a leaf.

 Butterfly caterpillar on a leaf.



 Butterfly chrysalis hanging from a twig.

 Adult butterfly on a flower.

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Adult butterfly

This cycle shows metamorphosis, the process of changing from one form to another.

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Frog

 Frog eggs in water.

Eggs



 Tadpole swimming in water.

Tadpole



 Froglet with legs and a tail.

Froglet



 Adult frog on a lily pad.

Adult frog

It begins in water. As the frog grows, legs develop and the tail becomes smaller.

Bean Plant

 Bean seed beginning to sprout.

Seed



 Bean seedling with first leaves.

Seedling



 Young bean plant growing taller.

Young plant

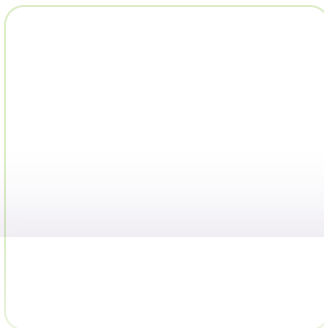


 Mature bean plant with flowers and bean pods.

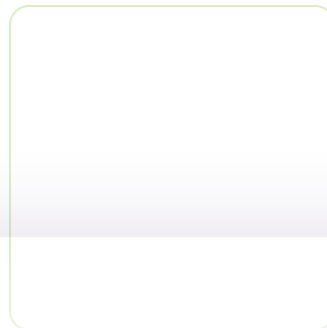
Mature plant

Plants have life cycles too. The mature plant makes flowers and seeds that start the pattern again.

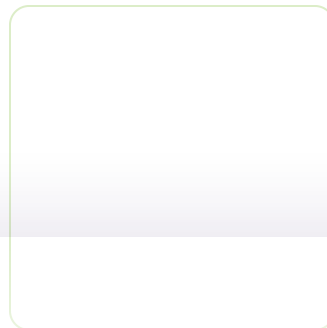
Human



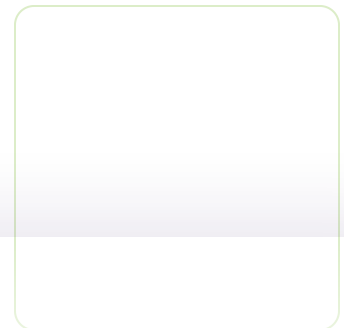
Beginning of life



Growth



Can reproduce



A new life begins

The adult does not turn back into a baby. The cycle continues when adults reproduce and a new baby begins life.



This is a preview. Unlock the full lesson.

Look for These Two Big Ideas

- Some organisms have life cycles that repeat over and over again.
- All organisms still follow the same broad pattern: birth, growth, reproduction, and death.
- Adults produce offspring that start the cycle in the first stage.

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Tracking Change Through One Life Cycle

Life Science — Life Cycles | Unit 1, Lesson 2 of 6

 **Grade:** Grade 3

 **Duration:** 45–60 minutes

 **Subject:** Life Science

FROM BROAD PATTERN TO EVIDENCE

In Lesson 1, students learned that all life cycles share the same big pattern: **birth, growth, reproduction, and death**. In this lesson, they track one organism across time and decide when the organism is still the same individual changing and when a *new generation* begins.

The key idea: growth changes one organism over time, but reproduction starts a new organism and keeps the cycle going.

NGSS Standard:

3-LS1-1 — Develop models to describe that organisms have unique and diverse life cycles but all have in common birth, growth, reproduction, and death.

The Big Idea

One organism can change a lot as it grows, but it is still the same organism. A life cycle becomes a *cycle* when an adult reproduces and a new young organism begins the pattern again.

The Science



Plant Life Cycles and Seeds

Life Science — Life Cycles | Unit 1, Lesson 3 of 6

 **Grade:** 3rd Grade

 **Duration:** 45–60 minutes

 **Subject:** Life Science — Life Cycles



Seeds, seedlings, flowers, and new seeds show a plant life cycle in motion.

NGSS Standard:

3-LS1-1 — Develop models to describe that organisms have unique and diverse life cycles but all have in common birth, growth, reproduction, and death.

Learning Objective

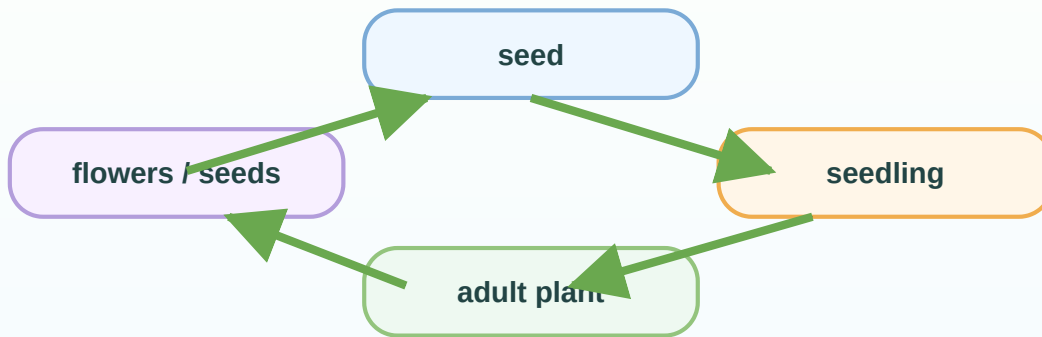
Students will model a plant life cycle and explain where birth, growth, reproduction, and death appear in that model.

The Big Idea

Plants may look different from animals, but they still follow the same big pattern of birth, growth, reproduction, and death. A seed begins life, a seedling grows, a mature plant reproduces by making new seeds, and the cycle can begin again.



SCIENCE SNAPSHOT



A plant life cycle shows how a seed can become a plant that produces new seeds.

What is happening?

Plant life cycles include stages such as seed, seedling, adult plant, flower, fruit, and new seeds. The exact timing varies, but the major scientific idea is that plants make offspring through seeds or other reproductive structures.

Why it works that way

Seeds protect the embryo and often store food for early growth. Flowers and fruits help with pollination, protection, and dispersal, which increases the chance that the next generation can grow successfully.

Look for this in the lesson

Students should connect each plant stage to a job, such as germination, growth, flowering, or seed making. That helps the cycle feel meaningful rather than like a list to memorize.

Ask while you read: Which stage in the plant cycle helps begin the next generation, and how?



Animal Life Cycles and Metamorphosis

Life Science — Life Cycles | Unit 1, Lesson 4 of 6

 **Grade:** 3rd Grade

 **Duration:** 45–60 minutes

 **Subject:** Life Science — Life Cycles



Some animals change gradually, and some change dramatically through metamorphosis.

NGSS Standard:

3-LS1-1 — Develop models to describe that organisms have unique and diverse life cycles but all have in common birth, growth, reproduction, and death.

Learning Objective

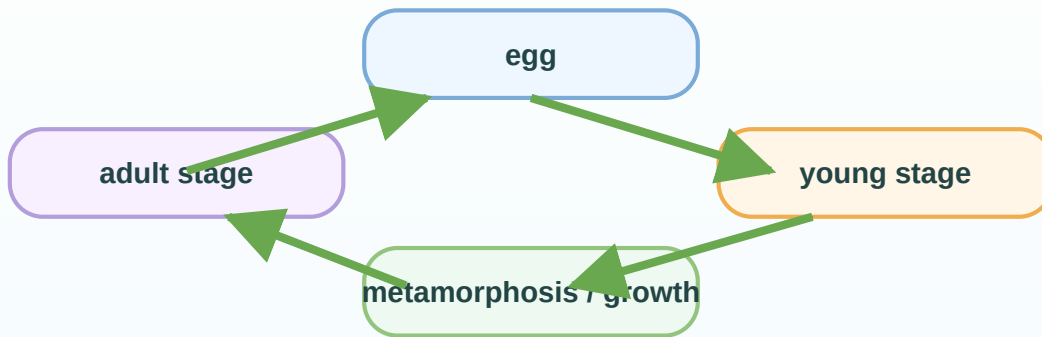
Students will compare animal life cycles and explain how metamorphosis and gradual growth can both fit the same shared life-cycle pattern.

The Big Idea

Animal life cycles can look very different from one another. Some animals hatch from eggs, some are born live, and some go through metamorphosis, which means their bodies change a lot as they grow. Even with those big differences, animal life cycles still follow the same big pattern.



SCIENCE SNAPSHOT



Metamorphosis shows that life-cycle stages can have very different bodies and jobs.

What is happening?

Animal life cycles can include gradual growth or dramatic metamorphosis. In metamorphosis, the body changes substantially between stages because different stages can solve different survival problems.

Why it works that way

A caterpillar is specialized for eating and growing, while an adult butterfly is specialized for flight and reproduction. Tadpoles and frogs also show how structure and habitat can change across a life cycle.

Look for this in the lesson

Students should notice that a dramatic change does not mean the organism became something unrelated. It is still the same organism moving through its life cycle.

Ask while you read: Why might it help an organism to have different body forms at different life stages?



Comparing Life Cycle Patterns

Life Science — Life Cycles | Unit 1, Lesson 5 of 6

 **Grade:** 3rd Grade

 **Duration:** 45–60 minutes

 **Subject:** Life Science — Life Cycles



Scientists compare patterns to explain what is shared and what is unique.

 NGSS Standard:

3-LS1-1 — Develop models to describe that organisms have unique and diverse life cycles but all have in common birth, growth, reproduction, and death.

Learning Objective

Students will compare plant and animal life-cycle models and use evidence to explain what stays the same and what changes across organisms.

The Big Idea

Scientists do not just memorize stage names. They compare models to look for patterns and explain with evidence. In this lesson, students compare plant and animal life cycles more directly so they can explain what stays the same across very different organisms.



SCIENCE SNAPSHOT

Shared life-cycle pattern

- begin
- grow
- reproduce

compare

- - - - -

Different details

- stage names
- body changes
- timing / habitat

Comparisons help students see the shared pattern behind different kinds of life cycles.

What is happening?

Comparing life cycles helps scientists see both similarities and differences across organisms. Plants and animals may have different stages, but both show growth, development, and reproduction.

Why it works that way

A comparison is strongest when students look for patterns in function, not only appearance. One stage may focus on growth, another on dispersal, and another on reproduction even if the organism looks very different from another species.

Look for this in the lesson

Students should practice saying what is the same at a big-pattern level and what is different in the stage details. That is what makes comparison more scientific than just noticing surface features.

Ask while you read: What is one important similarity and one meaningful difference between the cycles you compared?



Model and Explain a Life Cycle

Life Science — Life Cycles | Unit 1, Lesson 6 of 6

 **Grade:** 3rd Grade

 **Duration:** 45–60 minutes

 **Subject:** Life Science — Life Cycles



A final model should show stage order and explain the shared pattern clearly.

NGSS Standard:

3-LS1-1 — Develop models to describe that organisms have unique and diverse life cycles but all have in common birth, growth, reproduction, and death.

Learning Objective

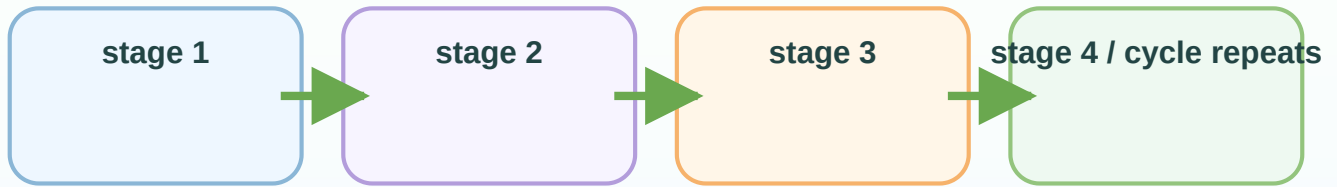
Students will create a life-cycle model for one organism and use that model as evidence to explain birth, growth, reproduction, and death.

The Big Idea

A scientific model is not just a neat drawing. It is a tool for showing relationships and explaining ideas. In this culminating lesson, students create and explain a life-cycle model that shows the shared pattern and uses evidence from the unit.



SCIENCE SNAPSHOT



A strong model helps other people understand how the organism's stages connect in time and purpose.

What is happening?

A life-cycle model should show the sequence of stages and explain how each stage contributes to survival or reproduction. Good models simplify reality, but they still keep the key pattern clear.

Why it works that way

Scientists use models to communicate ideas that unfold over time. Arrows, labels, and notes about what happens at each stage make the model much more useful than a picture with no explanation.

Look for this in the lesson

Students should treat the model as an explanation tool. The model is strongest when it shows both order and function.

Ask while you read: Does your model explain what happens at each stage or only show what it looks like?